

GOOD TO KNOW

A short within a series-parallel circuit does not necessarily mean that the circuit will draw excessive current from the voltage source.

6–7 Troubleshooting: Opens and Shorts in Series-Parallel Circuits

A short circuit has practically zero resistance. Its effect, therefore, is to allow excessive current. An open circuit has the opposite effect because an open circuit has infinitely high resistance with practically zero current. Furthermore, in series-parallel circuits, an open or short circuit in one path changes the circuit for the other resistances. For example, in Fig. 6–13, the series-parallel circuit in Fig. 6–13a becomes a series circuit with only R_1 when there is a short circuit between terminals A and B. As an example of an open circuit, the series-parallel circuit in Fig. 6–14a becomes a series circuit with just R_1 and R_2 and when there is an open circuit between terminals C and D.

Effect of a Short Circuit

We can solve the series-parallel circuit in Fig. 6–13a to see the effect of the short circuit. For the normal circuit with S_1 open, R_2 and R_3 are in parallel. Although R_3 is drawn horizontally, both ends are across R_2 . The switch S_1 has no effect as a parallel branch here because it is open.

The combined resistance of the $80\text{-}\Omega$ R_2 in parallel with the $80\text{-}\Omega$ R_3 is equivalent to $40\text{ }\Omega$. This $40\text{ }\Omega$ for the bank resistance is in series with the $10\text{-}\Omega$ R_1 . Then R_T is $40 + 10 = 50\text{ }\Omega$.

In the main line, I_T is $100/50 = 2\text{ A}$. Then V_1 across the $10\text{-}\Omega$ R_1 in the main line is $2 \times 10 = 20\text{ V}$. The remaining 80 V is across R_2 and R_3 as a parallel bank. As a result, $V_2 = 80\text{ V}$ and $V_3 = 80\text{ V}$.

Now consider the effect of closing switch S_1 . A closed switch has zero resistance. Not only is R_2 short-circuited, but R_3 in the bank with R_2 is also short-circuited. The closed switch short-circuits everything connected between terminals A and B. The result is the series circuit shown in Fig. 6–13b.

Now the $10\text{-}\Omega$ R_1 is the only opposition to current. I equals V/R_1 , which is $100/10 = 10\text{ A}$. This 10 A flows through the closed switch, through R_1 , and back to the positive terminal of the voltage source. With 10 A through R_1 , instead of its normal 2 A , the excessive current can cause excessive heat in R_1 . There is no current through R_2 and R_3 , as they are short-circuited out of the path for current.

Figure 6–13 Effect of a short circuit with series-parallel connections. (a) Normal circuit with S_1 open. (b) Circuit with short between points A and B when S_1 is closed; now R_2 and R_3 are short-circuited.

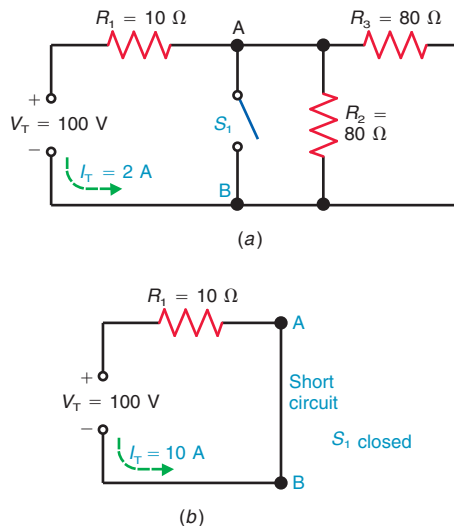
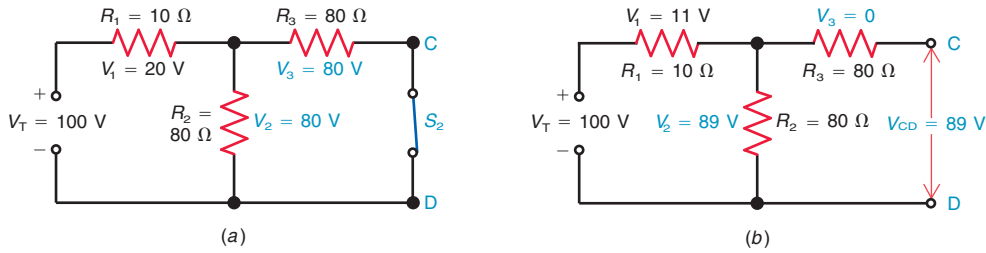


Figure 6–14 Effect of an open path in a series-parallel circuit. (a) Normal circuit with S_2 closed. (b) Series circuit with R_1 and R_2 when S_2 is open. Now R_3 in the open path has no current and zero IR voltage drop.



Effect of an Open Circuit

Figure 6–14a shows the same series-parallel circuit as Fig. 6–13a, except that switch S_2 is used now to connect R_3 in parallel with R_2 . With S_2 closed for normal operation, all currents and voltages have the values calculated for the series-parallel circuit. However, let us consider the effect of opening S_2 , as shown in Fig. 6–14b. An open switch has infinitely high resistance. Now there is an open circuit between terminals C and D. Furthermore, because R_3 is in the open path, its $80\ \Omega$ cannot be considered in parallel with R_2 .

The circuit with S_2 open in Fig. 6–14b is really the same as having only R_1 and R_2 in series with the 100-V source. The open path with R_3 has no effect as a parallel branch because no current flows through R_3 .

We can consider R_1 and R_2 in series as a voltage divider, where each IR drop is proportional to its resistance. The total series R is $80 + 10 = 90\ \Omega$. The $10\text{-}\Omega$ R_1 is $10/90$ or $1/9$ of the total R and the applied V_T . Then V_1 is $1/9 \times 100\text{ V} = 11\text{ V}$ and V_2 is $8/9 \times 100\text{ V} = 89\text{ V}$, approximately. The 11-V drop for V_1 and 89-V drop for V_2 add to equal the 100 V of the applied voltage.

Note that V_3 is zero. Without any current through R_3 , it cannot have any voltage drop.

Furthermore, the voltage across the open terminals C and D is the same 89 V as the potential difference V_2 across R_2 . Since there is no voltage drop across R_3 , terminal C has the same potential as the top terminal of R_2 . Terminal D is directly connected to the bottom end of resistor R_2 . Therefore, the potential difference from terminal C to terminal D is the same 89 V that appears across resistor R_2 .

Troubleshooting Procedures for Series-Parallel Circuits

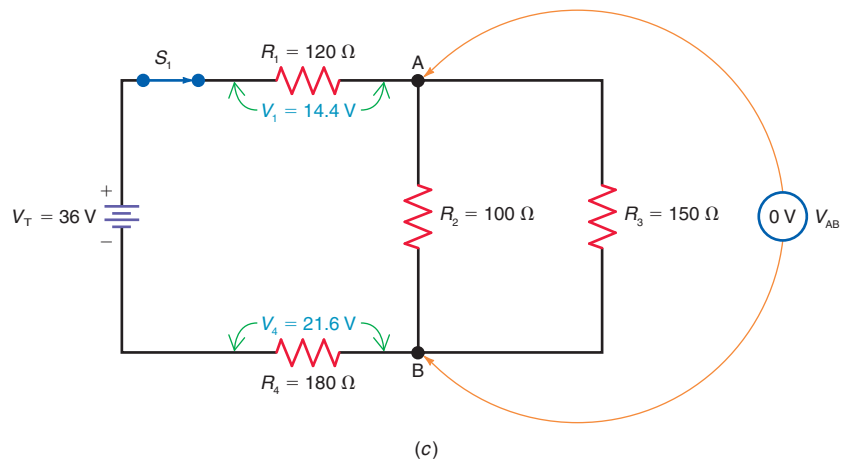
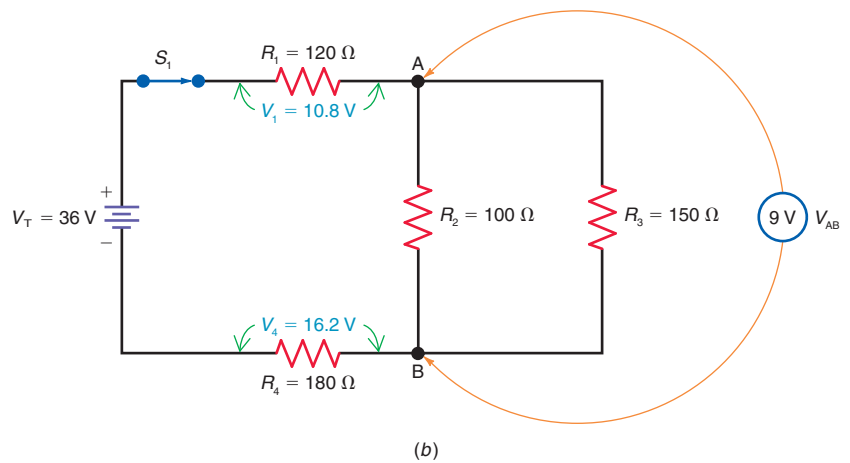
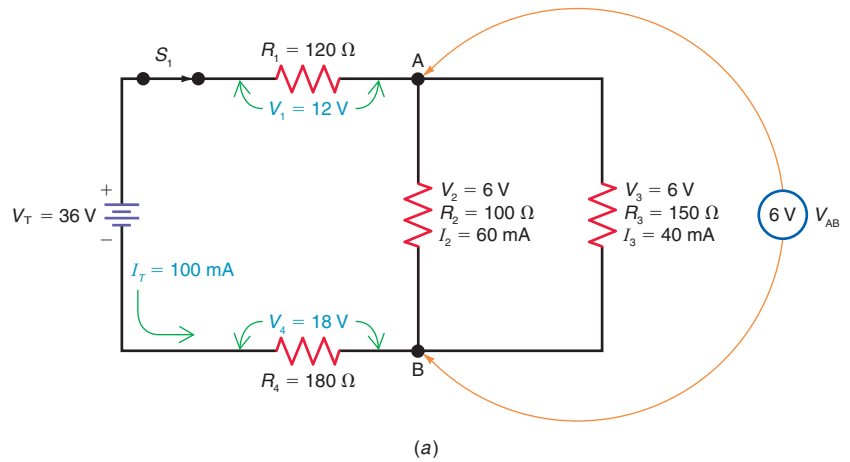
The procedure for troubleshooting series-parallel circuits containing opens and shorts is a combination of the procedures used to troubleshoot individual series and parallel circuits. Figure 6–15a shows a series-parallel circuit with its normal operating voltages and currents. Across points A and B, the equivalent resistance R_{EQ} of R_2 and R_3 in parallel is calculated as

$$\begin{aligned} R_{EQ} &= \frac{R_2 \times R_3}{R_2 + R_3} \\ &= \frac{100\ \Omega \times 150\ \Omega}{100\ \Omega + 150\ \Omega} \\ R_{EQ} &= 60\ \Omega \end{aligned}$$

GOOD TO KNOW

Troubleshooting a defective electronic circuit is like being a crime scene investigator. A technician gathers facts and data about the defective electronic circuit by making voltage, current, and resistance measurements. These measurements are then carefully analyzed to pinpoint the defective electronic component.

Figure 6-15 Series-parallel circuit for troubleshooting analysis. (a) Normal circuit voltages and currents; (b) circuit voltages with R_3 open between points A and B; (c) circuit voltages with R_2 or R_3 shorted between points A and B.



Since R_2 and R_3 are in parallel across points A and B, this equivalent resistance is designated R_{AB} . Therefore, $R_{AB} = 60\ \Omega$. The total resistance, R_T , is

$$\begin{aligned} R_T &= R_1 + R_{AB} + R_4 \\ &= 120\ \Omega + 60\ \Omega + 180\ \Omega \\ R_T &= 360\ \Omega \end{aligned}$$

The total current, I_T , is:

$$\begin{aligned} I_T &= \frac{V_T}{R_T} \\ &= \frac{36\ \text{V}}{360\ \Omega} \end{aligned}$$

$$I_T = 100\ \text{mA}$$

The voltage drops across the individual resistors are calculated as

$$\begin{aligned} V_1 &= I_T \times R_1 \\ &= 100\ \text{mA} \times 120\ \Omega \\ V_1 &= 12\ \text{V} \\ V_2 &= V_3 = V_{AB} = I_T \times R_{AB} \\ &= 100\ \text{mA} \times 60\ \Omega \\ &= 6\ \text{V} \\ V_4 &= I_T \times R_4 \\ &= 100\ \text{mA} \times 180\ \Omega \\ &= 18\ \text{V} \end{aligned}$$

The current in resistors R_2 and R_3 across points A and B can be found as follows:

$$\begin{aligned} I_2 &= \frac{V_{AB}}{R_2} \\ &= \frac{6\ \text{V}}{100\ \Omega} \\ I_2 &= 60\ \text{mA} \\ I_3 &= \frac{V_{AB}}{R_3} \\ &= \frac{6\ \text{V}}{150\ \Omega} \\ I_3 &= 40\ \text{mA} \end{aligned}$$

Example 6-3

Assume that the series-parallel circuit in Fig. 6-15a has failed. A technician troubleshooting the circuit has measured the following voltages:

$$\begin{aligned} V_1 &= 10.8\ \text{V} \\ V_{AB} &= 9\ \text{V} \\ V_4 &= 16.2\ \text{V} \end{aligned}$$

These voltage readings are shown in Fig. 6-15b. Based on the voltmeter readings shown, which component is defective and what type of defect does it have?

18. In Fig. 6–17, assume that the standard resistor, R_s , has been adjusted so that the current in M_1 is exactly $0\ \mu\text{A}$. How much voltage exists at terminal C with respect to terminal B?

- 1.1 V.
- 0 V.
- 10.9 V.
- none of the above.

19. In Fig. 6–17, assume that the ratio arm resistors, R_1 and R_2 , are

interchanged. What is the value of the unknown resistor, R_x , if R_s equals $33,950\ \Omega$ when the bridge is balanced?

- 339.5 k Ω .
- 3.395 k Ω .
- 33,950 Ω .
- none of the above.

20. In Fig. 6–17, assume that the standard resistor, R_s , cannot be adjusted high enough to provide a balanced condition. What

modification must be made to the circuit?

- Change the ratio arm fraction R_1/R_2 , from $1/10$ to $1/100$ or something less.
- Change the ratio arm fraction, R_1/R_2 from $1/10$ to $1/1$, $10/1$ or something greater.
- Reverse the polarity of the applied voltage, V_T .
- None of the above.

Essay Questions

- In a series-parallel circuit, how can you tell which resistances are in series and which are in parallel?
- Draw a schematic diagram showing two resistances in a parallel bank that is in series with one resistance.
- Draw a diagram showing how to connect three resistances of equal value so that the combined resistance will be $1\frac{1}{2}$ times the resistance of one unit.
- Draw a diagram showing two strings in parallel across a voltage source, where each string has three series resistances.
- Explain why components are connected in series-parallel, showing a circuit as an example of your explanation.
- Give two differences between a short circuit and an open circuit.
- Explain the difference between voltage division and current division.
- In Fig. 6–12, assume that the thermistor has a positive temperature coefficient (PTC). Explain what happens to the voltage V_{CD} (a) if the ambient temperature decreases; (b) if the ambient temperature increases.
- Draw a circuit with nine 40-V, 100-W bulbs connected to a 120-V source.
- (a) Two 10- Ω resistors are in series with a 100-V source. If a third 10- Ω R is added in series, explain why I will decrease. (b) The same two 10- Ω resistors are in parallel with the 100-V source. If a third 10- Ω R is added in parallel, explain why I_T will increase.

Problems

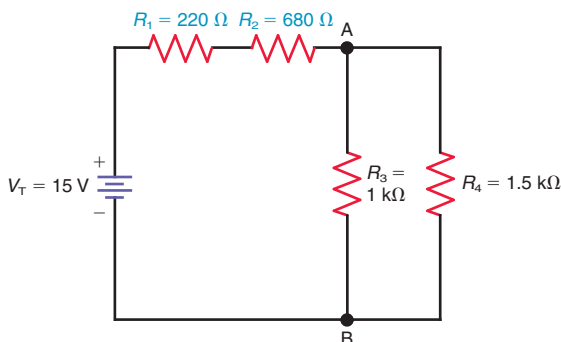
SECTION 6–1 FINDING R_T FOR SERIES-PARALLEL RESISTANCES

6–1 In Fig. 6–18, identify which components are in series and which ones are in parallel.

6–2 In Fig. 6–18,

- How much is the total resistance of just R_1 and R_2 ?
- What is the equivalent resistance of R_3 and R_4 across points A and B?
- How much is the total resistance, R_T , of the entire circuit?
- How much is the total current, I_T , in the circuit?
- How much current flows into point B?
- How much current flows away from point A?

Figure 6–18



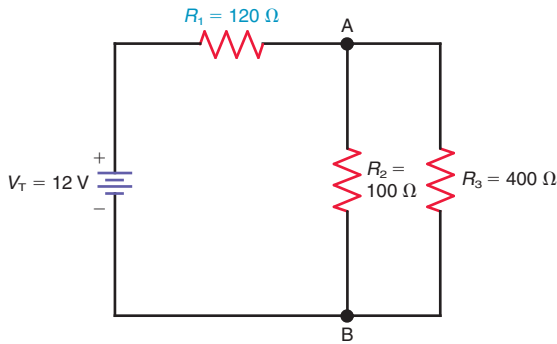
6–3 **MultiSim** In Fig. 6–18, solve for the following: I_1 , I_2 , V_1 , V_2 , V_3 , V_4 , I_3 , and I_4 .

6–4 In Fig. 6–19, identify which components are in series and which ones are in parallel.

6–5 In Fig. 6–19,

- What is the equivalent resistance of R_2 and R_3 across points A and B?
- How much is the total resistance, R_T of the entire circuit?
- How much is the total current, I_T , in the circuit?
- How much current flows into point B and away from point A?

Figure 6-19

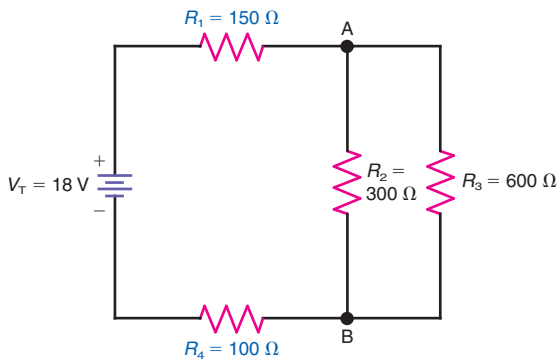


6-6 **MultiSim** In Fig. 6-19, solve for I_1 , V_1 , V_2 , V_3 , I_2 , and I_3 .

6-7 In Fig. 6-19, solve for P_1 , P_2 , P_3 , and P_T .

6-8 In Fig. 6-20, identify which components are in series and which ones are in parallel.

Figure 6-20



- 6-9 In Fig. 6-20,
- How much is the total resistance of just R_1 and R_4 ?
 - What is the equivalent resistance of R_2 and R_3 across points A and B?
 - How much is the total resistance, R_T , of the entire circuit?
 - How much is the total current, I_T , in the circuit?
 - How much current flows into point B and away from point A?

6-10 In Fig. 6-20, solve for the following: I_1 , V_1 , V_2 , V_3 , I_2 , I_3 , I_4 , V_4 , P_1 , P_2 , P_3 , P_4 , and P_T .

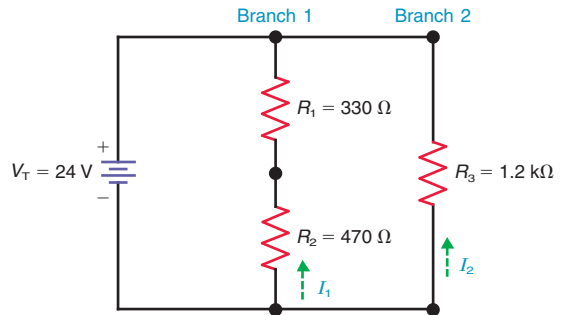
SECTION 6-2 RESISTANCE STRINGS IN PARALLEL

6-11 In Fig. 6-21,

- What is the total resistance of branch 1?
- What is the resistance of branch 2?
- How much are the branch currents I_1 and I_2 ?

- How much is the total current, I_T , in the circuit?
- How much is the total resistance, R_T , of the entire circuit?
- What are the values of V_1 , V_2 , and V_3 ?

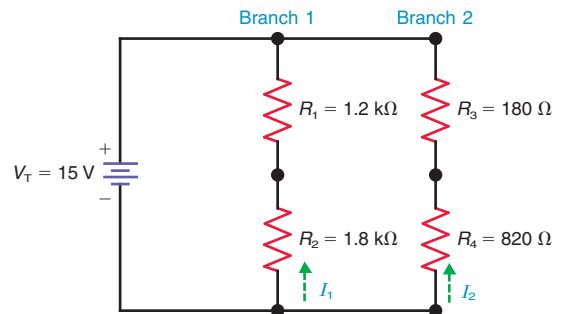
Figure 6-21



6-12 In Fig. 6-22,

- What is the total resistance of branch 1?
- What is the total resistance of branch 2?
- How much are the branch currents I_1 and I_2 ?
- How much is the total current, I_T , in the circuit?
- How much is the total resistance, R_T , of the entire circuit?
- What are the values of V_1 , V_2 , V_3 , and V_4 ?

Figure 6-22



6-13 In Fig. 6-23,

- What is the total resistance of branch 1?
- What is the total resistance of branch 2?
- How much are the branch currents I_1 and I_2 ?
- How much is the total current, I_T , in the circuit?
- How much is the total resistance, R_T , of the entire circuit?
- What are the values of V_1 , V_2 , V_3 , and V_4 ?

6-14 In Fig. 6-24, solve for

- branch currents I_1 , I_2 and the total current, I_T .
- R_T .
- V_1 , V_2 , V_3 , and V_4 .
- P_1 , P_2 , P_3 , P_4 and P_T .

Figure 6–23

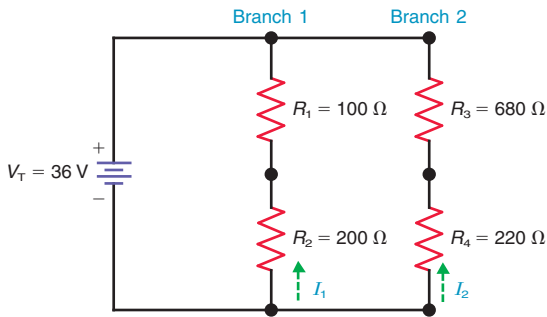
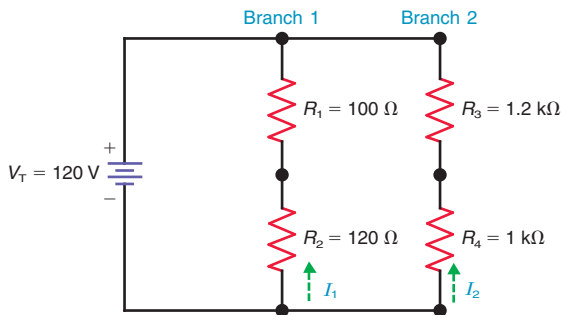


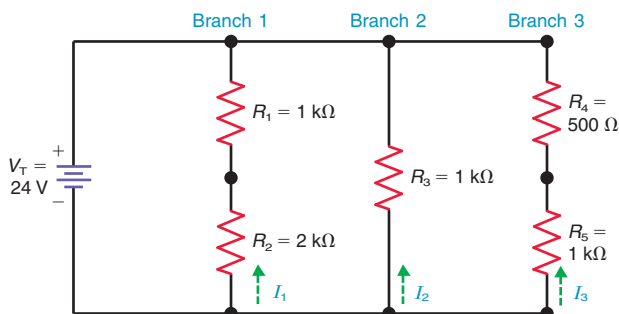
Figure 6–24



6–15 In Fig. 6–25, solve for

- branch currents I_1 , I_2 , I_3 and the total current, I_T .
- R_T .
- V_1 , V_2 , V_3 , V_4 , and V_5 .

Figure 6–25



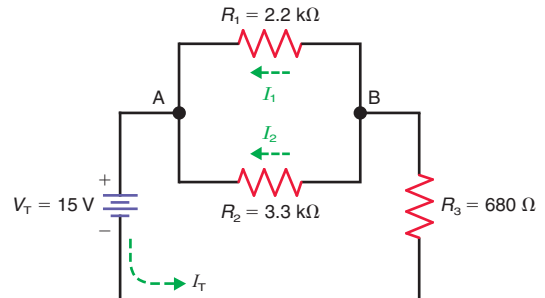
SECTION 6–3 RESISTANCE BANKS IN SERIES

6–16 In Fig. 6–26,

- What is the equivalent resistance of R_1 and R_2 in parallel across points A and B?
- What is the total resistance, R_T , of the circuit?
- What is the total current, I_T , in the circuit?

- How much voltage exists across points A and B?
- How much voltage is dropped across R_3 ?
- Solve for I_1 and I_2 .
- How much current flows into point B and away from point A?

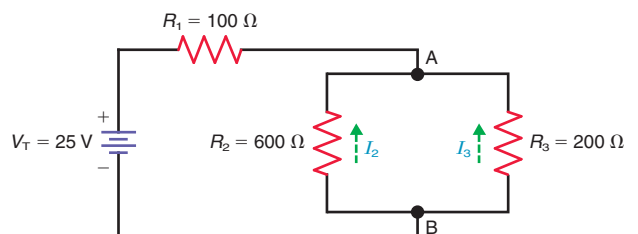
Figure 6–26



6–17 In Fig. 6–27,

- What is the equivalent resistance of R_2 and R_3 in parallel across points A and B?
- What is the total resistance, R_T , of the circuit?
- What is the total current, I_T , in the circuit?
- How much voltage exists across points A and B?
- How much voltage is dropped across R_1 ?
- Solve for I_2 and I_3 .
- How much current flows into point B and away from point A?

Figure 6–27



6–18 In Fig. 6–28, solve for

- R_T .
- I_T .
- V_1 , V_2 , and V_3 .
- I_1 and I_2 .

6–19 In Fig. 6–29, solve for

- R_T .
- I_T .
- V_1 , V_2 , and V_3 .
- I_2 and I_3 .

Figure 6-28

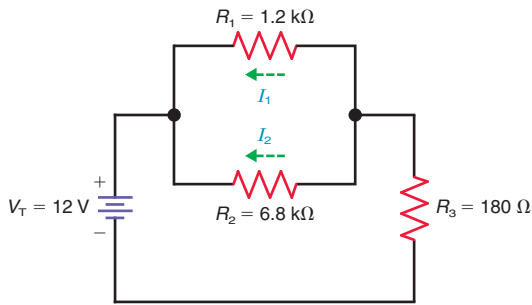
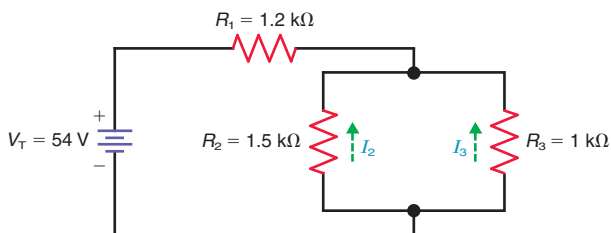


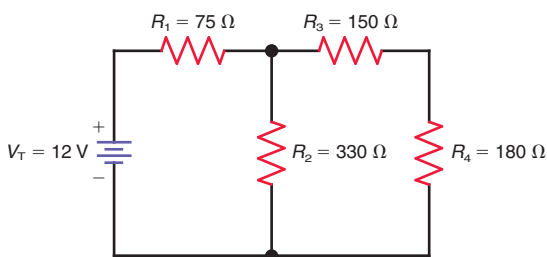
Figure 6-29



SECTION 6-4 RESISTANCE BANKS AND STRINGS IN SERIES-PARALLEL

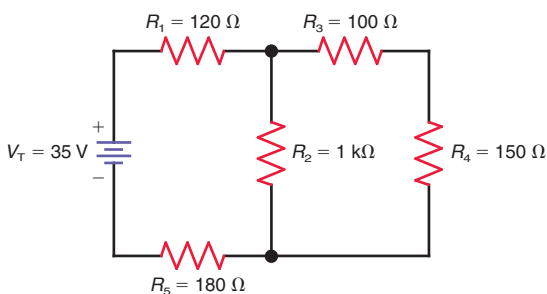
6-20 In Fig. 6-30, solve for R_T , I_T , V_1 , V_2 , V_3 , V_4 , I_1 , I_2 , I_3 , and I_4 .

Figure 6-30



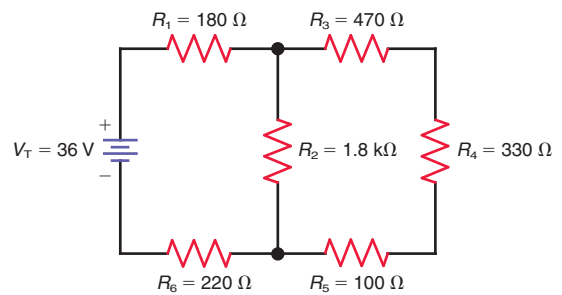
6-21 In Fig. 6-31, solve for R_T , I_T , V_1 , V_2 , V_3 , V_4 , V_5 , I_1 , I_2 , I_3 , I_4 , and I_5 .

Figure 6-31



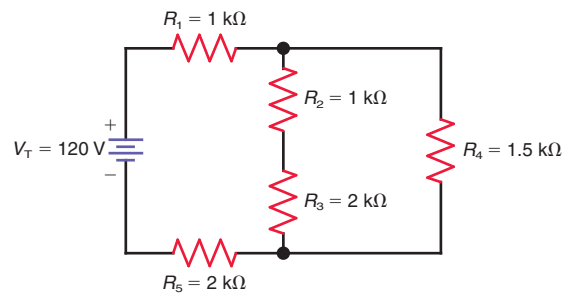
6-22 In Fig. 6-32, solve for R_T , I_T , V_1 , V_2 , V_3 , V_4 , V_5 , V_6 , I_1 , I_2 , I_3 , I_4 , I_5 , and I_6 .

Figure 6-32



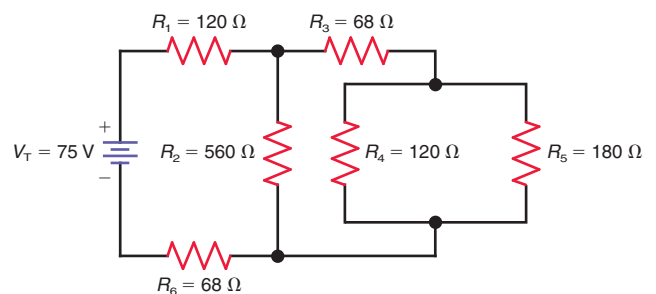
6-23 In Fig. 6-33, solve for R_T , I_T , V_1 , V_2 , V_3 , V_4 , V_5 , I_1 , I_2 , I_3 , I_4 , and I_5 .

Figure 6-33



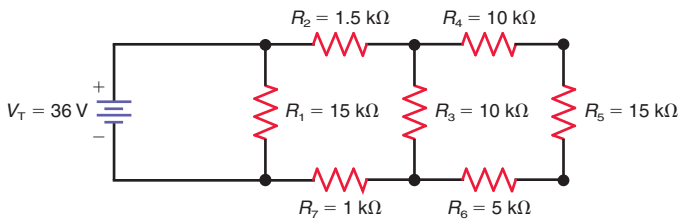
6-24 In Fig. 6-34, solve for R_T , I_T , V_1 , V_2 , V_3 , V_4 , V_5 , V_6 , I_1 , I_2 , I_3 , I_4 , I_5 , and I_6 .

Figure 6-34



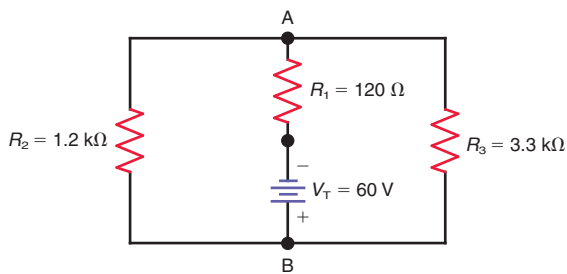
6-25 In Fig. 6-35, solve for R_T , I_T , V_1 , V_2 , V_3 , V_4 , V_5 , V_6 , V_7 , I_1 , I_2 , I_3 , I_4 , I_5 , I_6 , and I_7 .

Figure 6-35



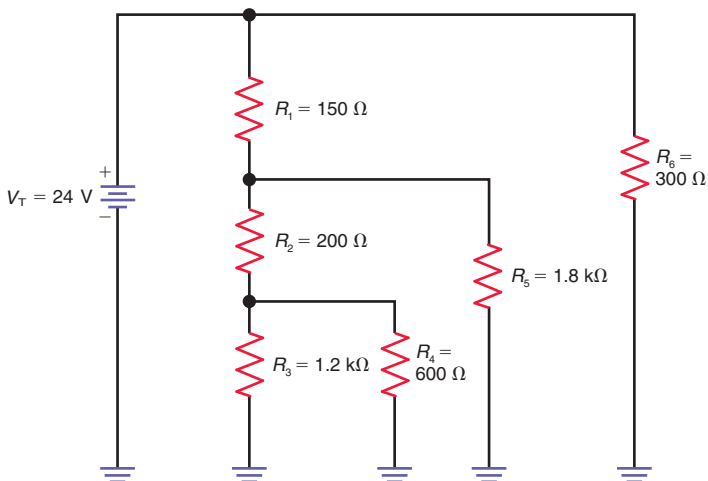
6-26 In Fig. 6-36, solve for R_T , I_T , I_2 , I_3 , V_1 , V_2 , V_3 , and the voltage, V_{AB} .

Figure 6-36



6-27 In Fig. 6-37, solve for R_T , I_T , V_1 , V_2 , V_3 , V_4 , V_5 , V_6 , I_1 , I_2 , I_3 , I_4 , I_5 , and I_6 .

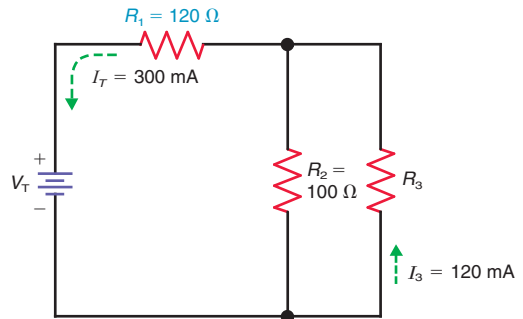
Figure 6-37



SECTION 6-5 ANALYZING SERIES-PARALLEL CIRCUITS WITH RANDOM UNKNOWNNS

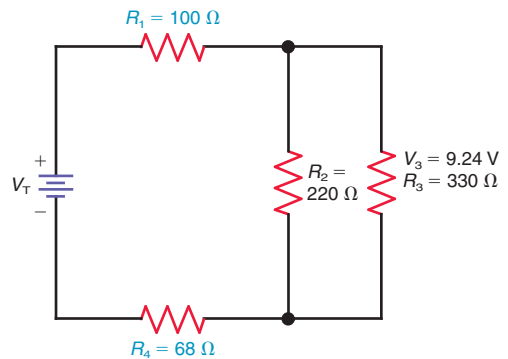
6-28 In Fig. 6-38, solve for V_T , V_2 , V_3 , I_2 , R_3 , R_T , and V_T .

Figure 6-38



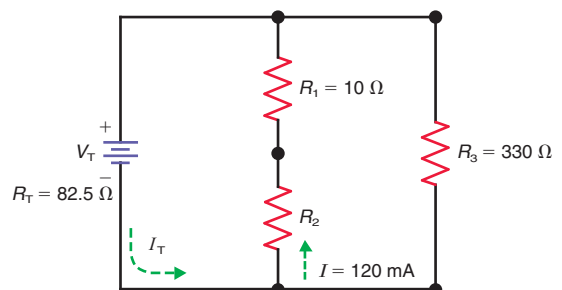
6-29 In Fig. 6-39, solve for R_T , I_T , V_T , V_1 , V_2 , V_4 , I_2 , and I_3 .

Figure 6-39



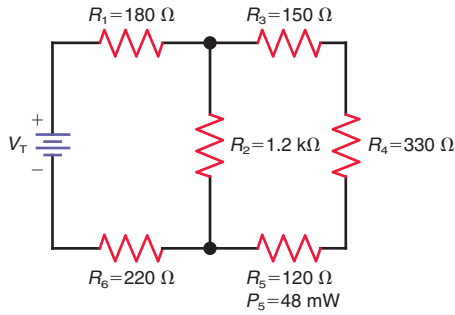
6-30 In Fig. 6-40, solve for R_2 , V_1 , V_2 , V_3 , I_3 , I_T , and V_T .

Figure 6-40



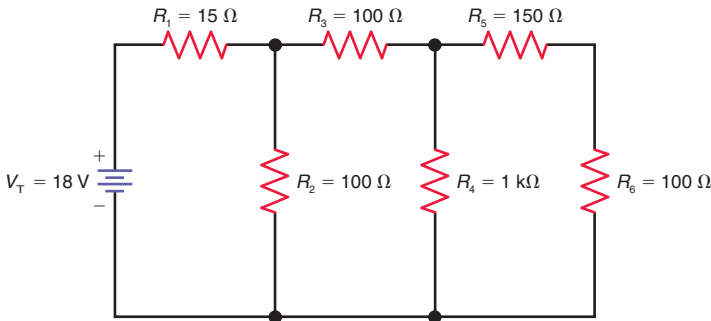
6-31 In Fig. 6-41, solve for R_T , I_T , V_1 , V_2 , V_3 , V_4 , V_5 , V_6 , I_2 , I_3 , I_4 , and I_5 .

Figure 6-41



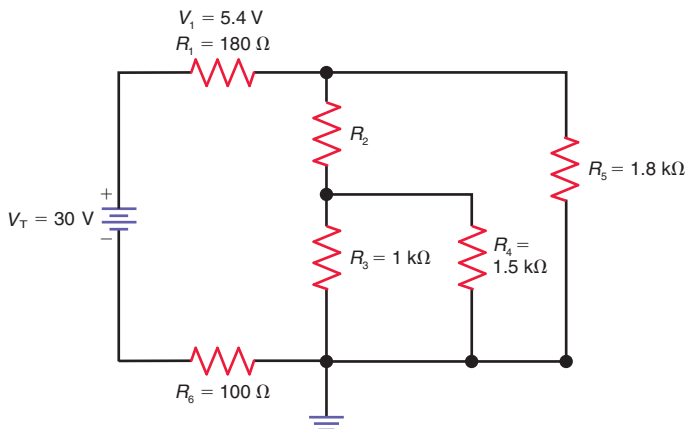
6-32 In Fig. 6-42, solve for R_T , I_T , V_1 , V_2 , V_3 , V_4 , V_5 , V_6 , I_2 , I_3 , I_4 , I_5 , and I_6 .

Figure 6-42



6-33 In Fig. 6-43, solve for I_T , R_T , R_2 , V_2 , V_3 , V_4 , V_5 , V_6 , I_2 , I_3 , I_4 , I_5 , and I_6 .

Figure 6-43



SECTION 6-6 THE WHEATSTONE BRIDGE

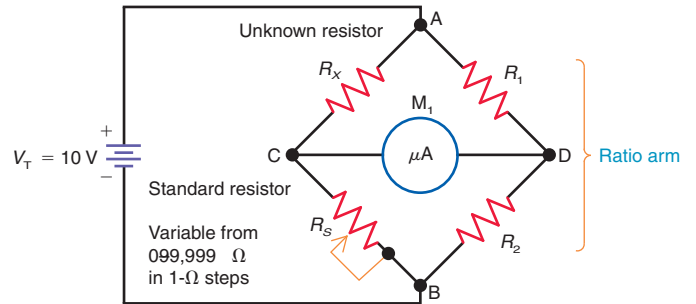
Probs. 34-38 refer to Fig. 6-44.

6-34 In Fig. 6-44,

- How much current flows through M_1 when the Wheatstone bridge is balanced?

- How much voltage exists between points C and D when the bridge is balanced?

Figure 6-44



6-35 In Fig. 6-44, assume that the bridge is balanced when $R_1 = 1 \text{ k}\Omega$, $R_2 = 5 \text{ k}\Omega$, and $R_S = 34,080 \Omega$. Determine

- the value of the unknown resistor, R_X .
- the voltages V_{CB} and V_{DB} .
- the total current, I_T , flowing to and from the voltage source, V_T .

6-36 In reference to Prob. 6-35, which direction (C to D or D to C) will electrons flow through M_1 if

- R_S is reduced in value?
- R_S is increased in value?

6-37 In Fig. 6-44, calculate the maximum unknown resistor, $R_{X(\max)}$, that can be measured for the following ratio arm values:

- $\frac{R_1}{R_2} = \frac{1}{1000}$
- $\frac{R_1}{R_2} = \frac{1}{100}$
- $\frac{R_1}{R_2} = \frac{1}{10}$
- $\frac{R_1}{R_2} = \frac{1}{1}$
- $\frac{R_1}{R_2} = \frac{10}{1}$
- $\frac{R_1}{R_2} = \frac{100}{1}$

6-38 Assume that the same unknown resistor, R_X , is measured using different ratio arm fractions in Fig. 6-44. In each case, the standard resistor, R_S , is adjusted to provide the balanced condition. The values for each measurement are

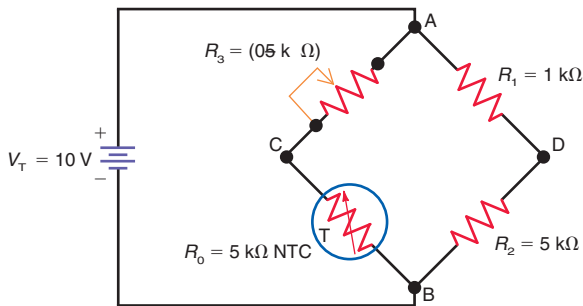
- $R_S = 123 \Omega$ and $\frac{R_1}{R_2} = \frac{1}{1}$.
- $R_S = 1232 \Omega$ and $\frac{R_1}{R_2} = \frac{1}{10}$.
- $R_S = 12,317 \Omega$ and $\frac{R_1}{R_2} = \frac{1}{100}$.

Calculate the value of the unknown resistor, R_X , for each measurement. Which ratio arm provides the greatest accuracy?

PROBLEMS 39–41 REFER TO FIG. 6–45.

6–39 In Fig. 6–45, to what value must R_3 be adjusted to provide zero volts across terminals C and D when the ambient temperature, T_A , is 25°C? (Note: R_0 is the resistance of the thermistor at an ambient temperature, T_A , of 25°C.)

Figure 6–45



6–40 In Fig. 6–45, assume that R_3 is adjusted to provide zero volts across terminals C and D at an ambient temperature, T_A , of 25°C. What happens to the polarity of the output voltage, V_{CD} , when

- the ambient temperature, T_A , increases above 25°C?
- the ambient temperature, T_A , decreases below 25°C?

6–41 In Fig. 6–45, assume that R_3 has been adjusted to 850 Ω to provide zero volts across the output terminals C and D. Determine

- the resistance of the thermistor.
- whether the ambient temperature, T_A , has increased or decreased from 25°C.

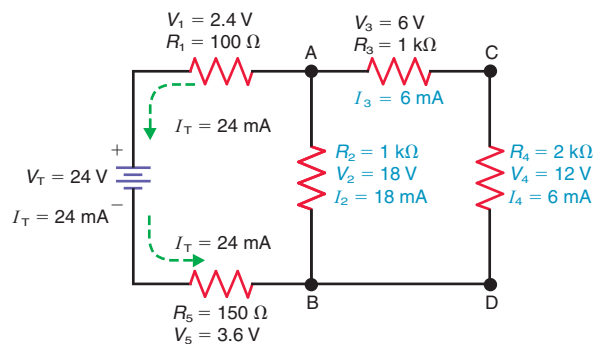
SECTION 6–7 TROUBLESHOOTING: OPENS AND SHORTS IN SERIES-PARALLEL CIRCUITS

Figure 6–46 shows a series-parallel circuit with its normal operating voltages and currents.

6–42 **MultiSim** In Fig. 6–46, determine the voltages V_1 , V_{AB} , V_3 , V_{CD} , and V_5 for each of the following component troubles:

- R_4 is open.
- R_2 is shorted.
- R_3 is open.
- R_4 is shorted.
- R_2 is open.
- R_1 is open.
- R_1 is shorted.

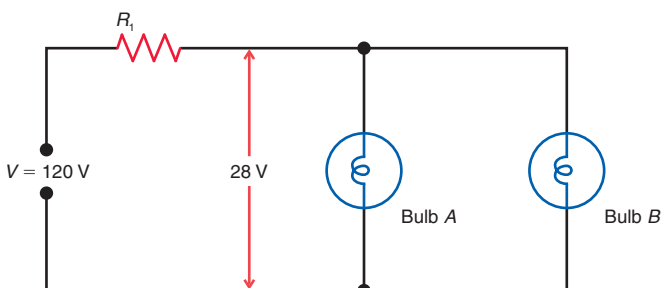
Figure 6–46



Critical Thinking

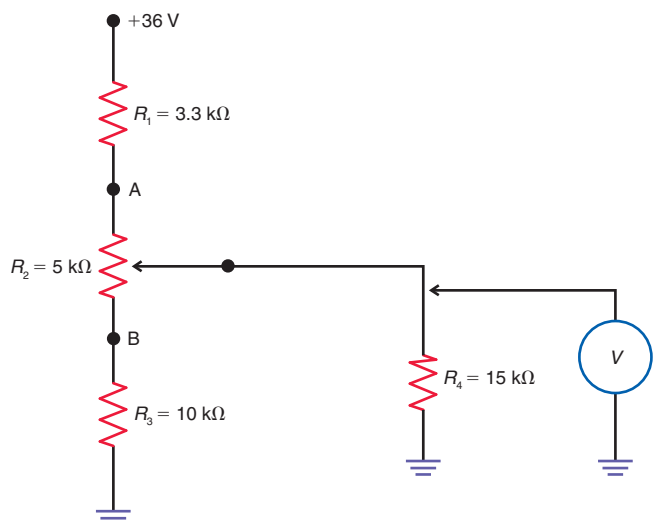
6–43 In Fig. 6–47, bulbs A and B each have an operating voltage of 28 V. If the wattage ratings for bulbs A and B are 1.12 W and 2.8 W, respectively, calculate (a) the required resistance of R_1 ; (b) the recommended wattage rating of R_1 ; (c) the total resistance R_T .

Figure 6–47 Circuit diagram for Critical Thinking Prob. 6–43.



6–44 Refer to Fig. 6–48. How much voltage will be indicated by the voltmeter when the wiper arm of the linear potentiometer R_2 is set (a) to point A; (b) to point B; (c) midway between points A and B?

Figure 6–48 Circuit diagram for Critical Thinking Prob. 6–44.



6–45 Explain how the temperature control circuit in Fig. 6–12 will be affected if the polarity of the applied voltage V_T is reversed.